

UNIVERSITY OF SCIENCE AND TECHNOLOGY OF SOUTHERN PHILIPPINES

CS 414 - 4B - ACTIVITY 1 (Part 1): Havil-Hakimi Algorithm

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Problem No. 5

A social network analyst wants to map interactions among 6 club members with the target degree sequence: $S_5 = (5, 4, 3, 2, 1, 3)$.

- a. Determining Whether the Sequence is Graphical

Preliminary Check:

Number of vertices (n) = 6

Sum of degrees = $5 + 4 + 3 + 2 + 1 + 3 = 18$ (even, necessary condition is satisfied)

Maximum degree = $5 = n - 1$ (does not exceed the maximum possible degree)

Applying the Hakimi Algorithm:

Step 1:

Sort in non-increasing order: $(5, 4, 3, 3, 2, 1)$

Remove the first element (5).

Subtract 1 from the next 5 elements:

$(4-1, 3-1, 3-1, 2-1, 1-1) = (3, 2, 2, 1, 0)$

No negative values. Proceed.

Step 2:

Sort in non-increasing order: $(3, 2, 2, 1, 0)$

Remove the first element (3).

Subtract 1 from the next 3 elements:

$(2-1, 2-1, 1-1, 0) = (1, 1, 0, 0)$

No negative values. Proceed.

Step 3:

Sort in non-increasing order: $(1, 1, 0, 0)$

Remove the first element (1).

Subtract 1 from the next 1 element:

$(1-1, 0, 0) = (0, 0, 0)$

No negative values. Proceed.

Step 4:

Remaining sequence: $(0, 0, 0)$

All elements are zero.

Conclusion: The sequence $S_5 = (5, 4, 3, 2, 1, 3)$ IS GRAPHICAL.
A simple graph with this exact degree sequence can be constructed.

b. Graph Construction

Vertex Assignment (sorted by degree):

$v_1 = \text{degree } 5$

$v_2 = \text{degree } 4$

$v_3 = \text{degree } 3$

$v_4 = \text{degree } 3$

$v_5 = \text{degree } 2$

$v_6 = \text{degree } 1$

Edge List (derived by tracing the algorithm):

From Step 1 (v_1 connects to all other vertices):

$v_1-v_2, v_1-v_3, v_1-v_4, v_1-v_5, v_1-v_6$

From Step 2 (v_2 connects to the next 3 highest-degree vertices):

$v_2-v_3, v_2-v_4, v_2-v_5$

From Step 3 (v_3 connects to the next 1 highest-degree vertex):

v_3-v_4

Total edges: 9

Adjacency Matrix:

	v_1	v_2	v_3	v_4	v_5	v_6
v_1	[0	1	1	1	1	1]
v_2	[1	0	1	1	1	0]
v_3	[1	1	0	1	0	0]
v_4	[1	1	1	0	0	0]
v_5	[1	1	0	0	0	0]
v_6	[1	0	0	0	0	0]

Verification:

v_1 : connected to $v_2, v_3, v_4, v_5, v_6 \rightarrow \text{degree } 5 \checkmark$

v_2 : connected to $v_1, v_3, v_4, v_5 \rightarrow \text{degree } 4 \checkmark$

v_3 : connected to $v_1, v_2, v_4 \rightarrow \text{degree } 3 \checkmark$

v_4 : connected to $v_1, v_2, v_3 \rightarrow \text{degree } 3 \checkmark$

v_5 : connected to $v_1, v_2 \rightarrow \text{degree } 2 \checkmark$

v_6 : connected to $v_1 \rightarrow \text{degree } 1 \checkmark$

Sum of degrees = $5+4+3+3+2+1 = 18 = 2 \times 9$ edges $\checkmark$

The graph satisfies the given degree sequence $S_5 = (5, 4, 3, 2, 1, 3)$.